Recognition in Computer Vision

Recognition Problems

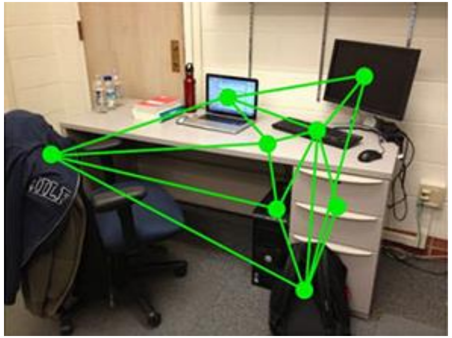
- What is it?
  - Object and scene recognition
- Who is it?
  - Identity recognition
- Where is it?
  - Object detection
- What are they doing?
  - Activities
- All of these are **classification** problems
  - Choose one class from a list of possible candidates

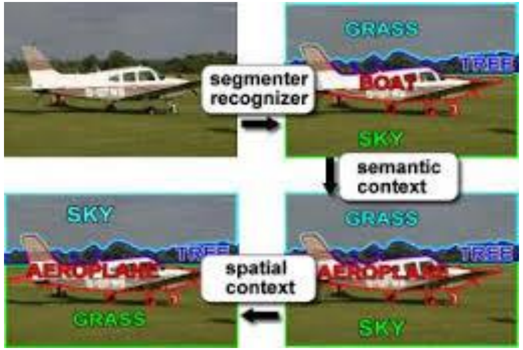
Types of Learning

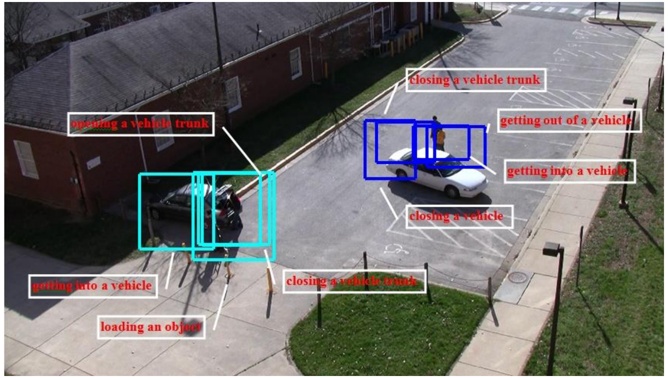
- **Supervised Learning** – learning from labeled data
- **Unsupervised Learning** – clustering, anomaly detection, dimensionality reduction
- **Self-supervised Learning** – supervisory signal from the structure of the data
- **Reinforcement Learning** – actions to maximize reward; state, action, reward
  - Deep RL: Policy(a|s) represented as a deep network
  - Imitation learning: expert provided demonstrations instead of reward function
- **Domain Adaptation** – domain shift or distribution shift between training and test data

Context in Visual Recognition

- In nature, objects and events tend to coexist with each other in a particular configuration, which is often termed as **context**.
- Context can provide significant visual clue for recognition.
- Recognition of one object/event can help recognizing others.

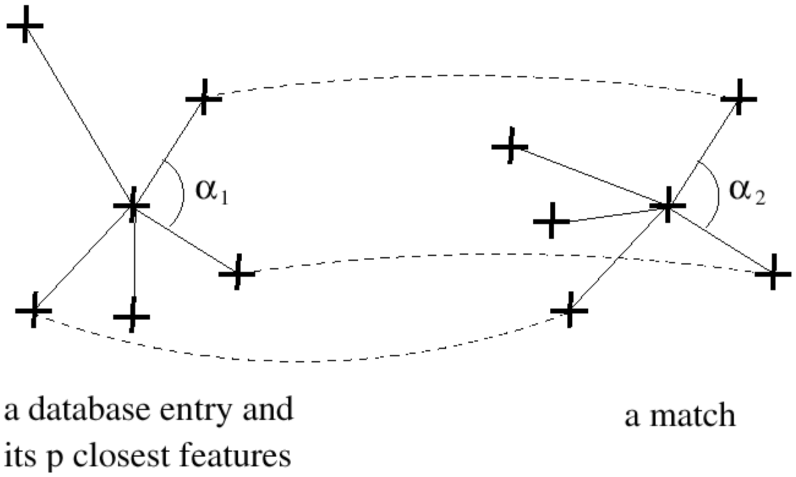




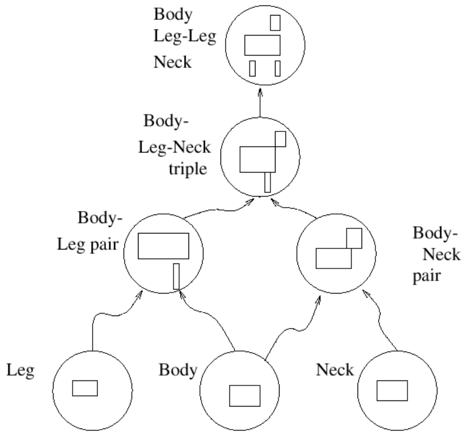


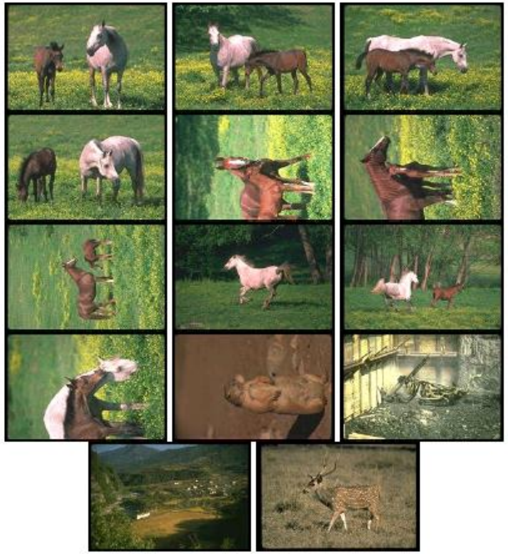
Matching Using Relationships

Employ spatial relations



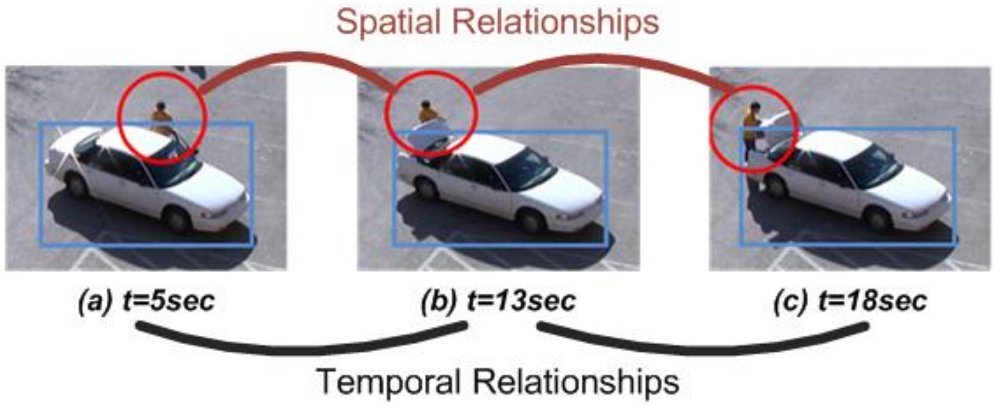
Recognition by parts



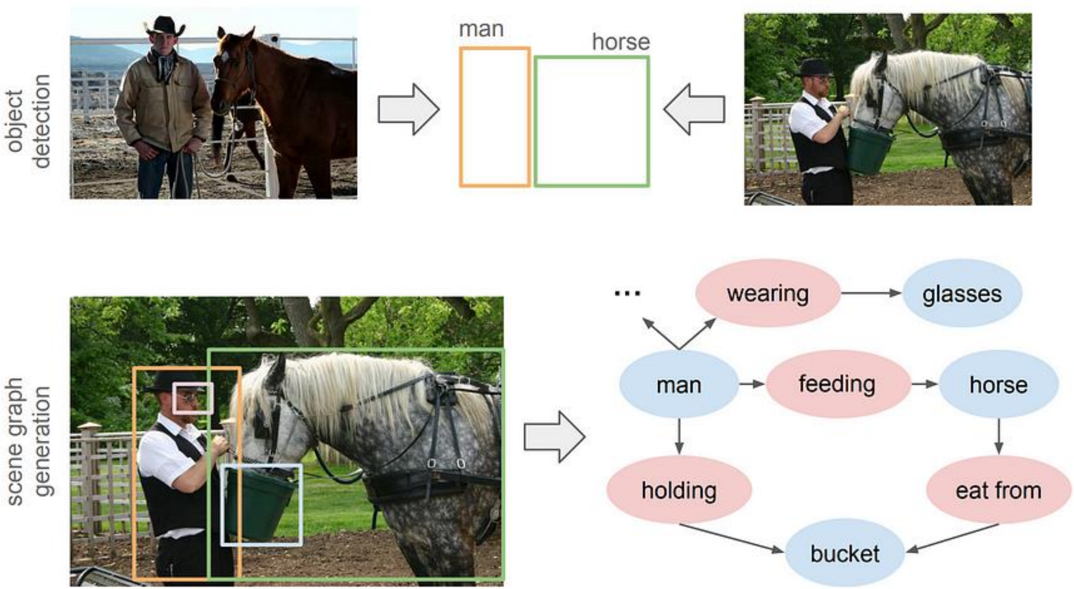


Computer Vision - A Modern Approach
Set: Recognition by relations
Slides by D.A. Forsyth

Spation-Temporal Relationships



Scene Graphs



https://medium.com/stanford-cs224w/scene-graph-generation-compression-and-classification-on-action-genome-dataset-9f692a1d5394

Dynamic Scene Graph

FRAME 1 | GROUND TRUTH | ESTIMATED SCENE GRAPH | FRAME 2 | GROUND TRUTH | ESTIMATED SCENE GRAPH